

■ Social Progress Analytics Engine

A professional-grade interactive analytics dashboard built with Streamlit that explores social progress, well-being, and country performance using World Happiness Report data.

!Python

!Streamlit

!Plotly

!License

■ Overview

The Social Progress Analytics Engine provides comprehensive analysis of global happiness and well-being data from the World Happiness Report (2011–2025). It enables users to:

- **Explore Rankings**: View and compare country rankings by life evaluation scores
- **Compare Countries**: Side-by-side analysis of multiple countries across all metrics
- **Analyze Factors**: Understand how GDP, social support, health, freedom, generosity, and corruption contribute to happiness
- **Track Trends**: Visualize temporal changes in scores, ranks, and factor contributions
- **Deep Dive**: Generate detailed country profiles with historical analysis and narrative insights
- **Export Data**: Download filtered datasets and summary reports

■ Getting Started

Prerequisites

- Python 3.8 or higher
- pip package manager

Installation

1. **Clone or download the repository**:

```
bash
```

```
cd social-progress-analytics
```

2. **Create a virtual environment** (recommended):

```
bash
```

```
python -m venv venv
```

```
# Windows
```

```
venv\Scripts\activate
# macOS/Linux
source venv/bin/activate
```

3. ****Install dependencies****:

```
bash
pip install -r requirements.txt
```

4. ****Run the application****:

```
bash
streamlit run app.py
```

5. Open your browser at <http://localhost:8501>

■ Project Structure

```
social-progress-analytics/
■■■■ README.md
■■■■ .gitignore
■■■■ requirements.txt
■■■■ app.py
■■■■ fix_keys.py
■■■■ data/
■ ■■■■ WHR26_Data_Figure_2.1.xlsx
■■■■ pages/
■ ■■■■ 1_Home.py
■ ■■■■ 2_Country_Rankings.py
■ ■■■■ 3_Country_Comparison.py
■ ■■■■ 4_Factor_Analysis.py
■ ■■■■ 5_Time_Trends.py
■ ■■■■ 6_Country_Profile.py
■ ■■■■ 7_Data_Table.py
■■■■ utils/
■ ■■■■ __init__.py
■ ■■■■ charts.py
■ ■■■■ config.py
■ ■■■■ data_loader.py
```

- notebook/
- ■■■■ Social_Progress_Analysis.ipynb
- screenshots/
- ■■■■ home.png
- ■■■■ country_rankings.png
- ■■■■ country_comparison.png
- ■■■■ factor_analysis.png
- ■■■■ time_trends.png
- ■■■■ country_profile.png
- ■■■■ data_table.png
- docs/
- Project_Report.pdf

■ **Dashboard Pages**

Page	Description
Home	Overview with KPI cards, top countries, and navigation
Country Rankings	Interactive ranking explorer with bar charts and confidence intervals
Country Comparison	Multi-country comparison with radar charts and trend lines
Factor Analysis	Correlation analysis, factor importance, and distribution views
Time Trends	Year-over-year changes, top movers, and global trends
Country Profile	Deep-dive single country analysis with narrative insights
Data Table	Filterable, sortable, downloadable data browser

■ **Dataset**

The application uses the **World Happiness Report 2026** dataset (WHR26_Data_Figure_2.1.xlsx) containing:

- 2,116 records** across **14 years** (2011–2025)
- Metrics**: Life evaluation (Cantril ladder), confidence intervals, and 7 explanatory factors
- Coverage**: 140+ countries ranked annually

Key Columns

Column	Description

Life Evaluation	3-year average Cantril ladder score (0-10)
Log GDP per capita	Economic contribution to happiness
Social support	Social network contribution
Healthy life expectancy	Health contribution
Freedom to make life choices	Freedom contribution
Generosity	Generosity contribution
Perceptions of corruption	Anti-corruption contribution
Dystopia + residual	Baseline + unexplained variance

■■ Deployment (Streamlit Cloud)

1. Push this repository to GitHub
2. Go to share.streamlit.io
3. Connect your GitHub repository
4. Set the main file path to `app.py`
5. Click **Deploy**

The app uses relative paths and is fully compatible with Streamlit Cloud.

■ Configuration

Key configuration options are in `utils/config.py`:

- **Color palette**: Customizable dark theme colors
- **Chart styling**: Plotly template and defaults
- **Factor mappings**: Short names to full column names
- **CSS theming**: Custom glassmorphism and gradient styles

■ Assumptions

- The dataset follows the World Happiness Report format
- Missing values in factor columns are handled gracefully (filtered out for relevant analyses)
- Life evaluation column is auto-detected by searching for 'life evaluation' in column names
- Years without data (e.g., 2013) are handled seamlessly

■ License

This project is for academic and portfolio use. The World Happiness Report data is publicly available.

Built with ❤️📊 using Streamlit, Plotly, and pandas.